

DEMO BIOFOULING HULL INSPECTION

SRI LANKA - DOCKYARD VESSEL



FRANMARINE

WEDNESDAY 24TH JUNE 2026



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EXECUTIVE SUMMARY

The example vessel, an 8,550 GT tanker, underwent a Biofouling Hull Inspection at Colombo Port on 24 June 2026. Inspection was carried out by ROV in reduced underwater visibility, with findings recorded in the MarineStream platform.

Inspection Findings

- **Running gear:** The propeller carried light, scattered macrofouling across all five blades (LoF Rank 2, 1–5%), with the boss clean (Rank 0). No antifouling coating was present on the propeller, consistent with a bare bronze finish.
- **Rudder:** The rudder presented the heaviest fouling on the vessel, with considerable macrofouling (LoF Rank 3, 6–15%) across the body, hinges, pintle bearings, flat plates, inspection ports and leading edge. Antifouling was worn from edges, seams and fixings to expose anticorrosion undercoat (PDR 20). Anodes were clean and coating-intact (Rank 1, PDR 10).
- **Sea chests:** Sea Chest 1 external face carried light scattered fouling (Rank 2) with minor antifouling loss and brush swirl marks in the outer coat (PDR 30). The internal sea chest was not visible; the grating was reported as fouled.

Niche Area Condition

- The rudder is the priority niche, combining the highest fouling level recorded and exposed undercoat at all movement and weld points.
- Sea Chest 1 internals could not be assessed and remain an outstanding item.

Key Concern

The rudder and propeller drive the management decision: considerable macrofouling on the rudder and light blade fouling represent the main contributors to drag and the highest reaccumulation risk. Coating breakdown on the rudder is confined to edges and fixings rather than the main plate, so the finding is fouling-led rather than coating-led.

Conclusion

No invasive marine species were identified during the inspection. The principal outstanding risk is the uninspected internal sea chest space, which could not be assessed by ROV in the available visibility and should be verified before any biosecurity sign-off.

1. PROJECT PARTICULARS

PROJECT:	Biofouling Hull Inspection	
VESSEL DETAILS:	Vessel:	Sri Lanka - Dockyard Vessel
	IMO:	
	Commissioned:	2008
	Gross Tonnage (t):	8550
	Length (m):	127 m
	Beam (m):	20 m
	Draft (m):	7.9 m
DATE OF INSPECTION:	Wednesday 24 June 2026	
CLIENT:	Sri Lanka Ports	
JOB LOCATION:	Colombo Port , Berth 8	
INSPECTION TEAM:	Supervisor:	Kyden Bradley
	ROV Unit:	Deep Trekker Pivot
	Inspector:	Mat Harvey



2. OVERVIEW

Franmarine was engaged by Sri Lanka Ports to conduct a Biofouling Hull Inspection on a example vessel at Colombo Port (Berth 8). The inspection was undertaken to assess the biofouling condition of the vessel's underwater surfaces, evaluate antifouling coating status, and identify any potential invasive marine species (IMS).

The scope of work and work method are based on the following legislation and guidelines: the Australian Biofouling Management Requirements (ABFMR), Version 2, 2023; the 2023 Guidelines for the control and management of ships' biofouling to minimise the transfer of invasive aquatic species (Resolution MEPC.378(80)); and the Craft Risk Management Standard (CRMS) for Biofouling on Vessels Arriving to New Zealand, issued by the Ministry of Primary Industries.

Franmarine has developed MarineStream, a biofouling management platform, to provide a secure and immutable record book that maintains compliance with the legislative and regulatory requirements listed above. Proactive biofouling management is the key to ensuring increased fuel efficiency through reduced drag, and to preventing the introduction of invasive marine species to home and foreign waters.



3. METHODOLOGY

Franmarine was directed to conduct a Biofouling Hull Inspection on a example vessel at Colombo Port, including the hull plating, sea chests, propeller, rudder, skeg, bilge keels, stern arrangement, draft marks and anodes. This inspection was performed to assess the current level of biofouling and identify any potential invasive marine species.

The inspection was conducted using a Deep Trekker Pivot ROV system, allowing for visual assessment of all accessible underwater surfaces from a mobile ROV system situated on the wharf.

Inspection findings were recorded in the MarineStream technology platform, ensuring inspection processes are standardised and an immutable record of the inspection findings is maintained. Biofouling was assessed using the Craft Risk Management Standard (CRMS) for Biofouling on Vessels Arriving to New Zealand as a guideline, specifically the Level of Fouling (LoF) definitions and typical examples. The condition of the antifouling coating was also assessed.

Documents used to assist the project include the Franmarine Operations Manual and client-supplied vessel drawings.

4. REFERENCE TABLES

The below tables are referenced throughout the report to formally standardise both biofouling and antifouling condition ratings referred to herein.

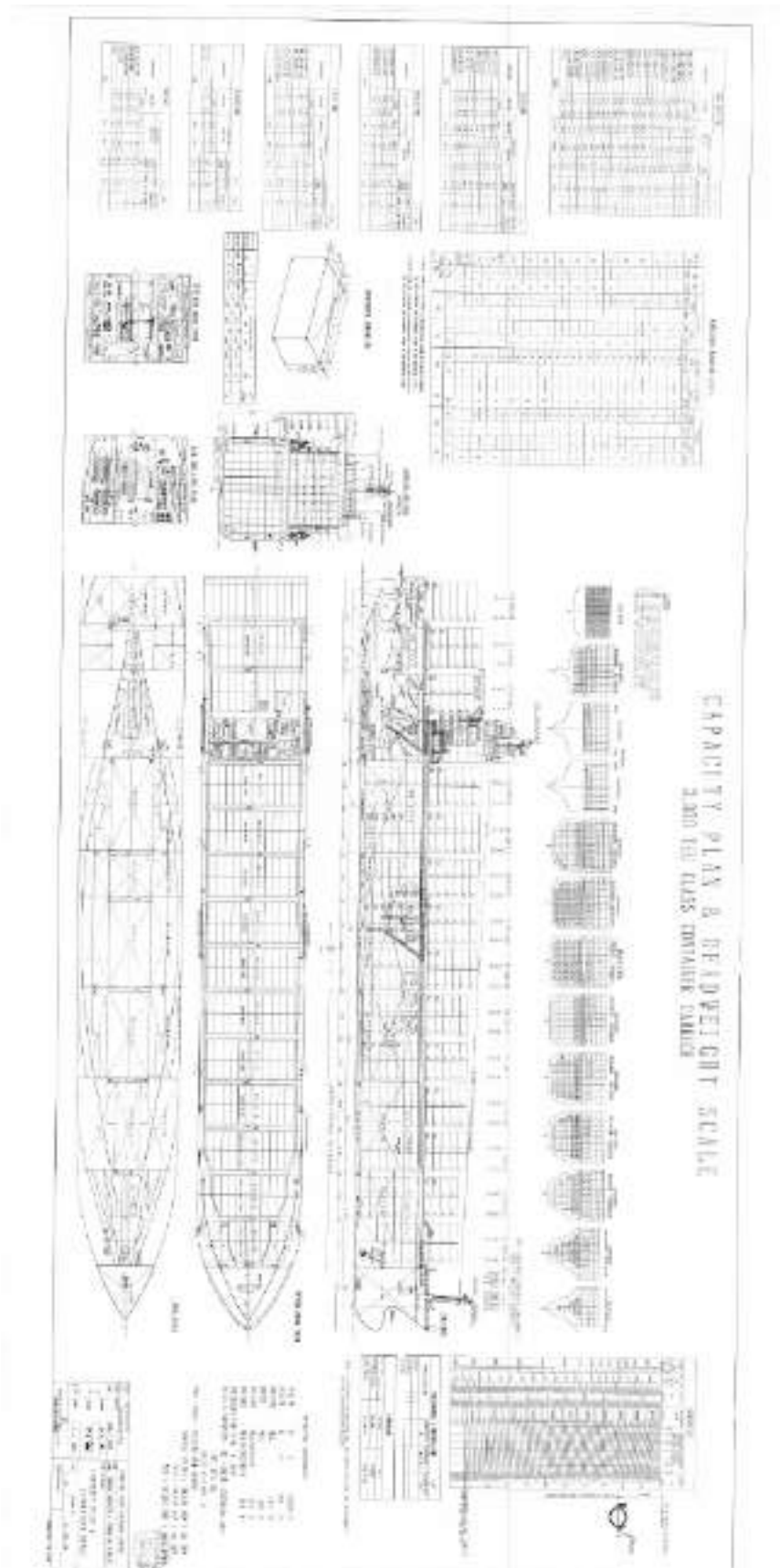
4.1 LEVEL OF FOULING (LOF) SCALE

LEVEL OF FOULING	DESCRIPTION
Rank: 0	0% No slime layer. No macrofouling. Only clean surfaces.
Rank: 1	0% Slime Layer on some or all surfaces. No macrofouling.
Rank: 2	1-5% of visible surfaces - Macrofouling present in small patches or a few isolated individuals or small colonies.
Rank: 3	6-15% of visible surfaces - Considerable macrofouling on surfaces.
Rank: 4	16-40% of visible surfaces - Extensive macrofouling present but more than half of surfaces without biofouling.
Rank: 5	41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.

4.2 PAINT DETERIORATION RATING (PDR) SCALE

RATING	DESCRIPTION
PDR: 0	No anti-foul (AF) coating applied.
PDR: 10	Anti-Foul (AF) Paint intact.
PDR: 20	AF paint missing from edges, corners, seams, welds, rivet, or bolt heads to expose anti corrosion (AC) paint. (undercoat).
PDR: 30	AF paint missing from slightly curved or flat areas to expose underlying AC paint; An AF paint with visible brush swirl marks within the outermost layer – not extending into underlying layers of paint.
PDR: 40	AF paint missing from intact blisters to expose AC paint or an AF coating with visible brush swirl marks exposing the next underlying layer of AF or AC paint.
PDR: 50	AF blisters ruptured to expose intact AC paint.
PDR: 60	AF/AC paint missing or peeling to expose steel substrate, no corrosion present.
PDR: 70	AF/AC paint removed from edges, corners, seams, welds, rivet or bolt heads to expose steel substrate with corrosion present.
PDR: 80	Ruptured AF/AC blisters on slightly curved or flat surfaces with corrosion or corrosion stains present.
PDR: 90	Area corrosion of steel substrate with no AF/AC paint cover due to peeling or abrasion damage.
PDR: 100	Area corrosion showing visible surface evidence of pitting, scaling, and roughening of steel substrate.

5. GENERAL ARRANGEMENT



5.1 RUDDER

5.1.1 OBSERVATIONS AND FINDINGS

Rudder inspected in poor visibility. Top of rudder, pintle bearings (port and starboard), rudder hinges (port and starboard), leading edge, rudder flat plates (port and starboard) and inspection ports (port and starboard) all exhibited considerable macrofouling (LoF Rank 3, 6–15% coverage), with antifouling missing from edges, corners, seams, welds and fixings to expose anticorrosion undercoat (PDR 20). Anodes clean (Rank 1) with antifouling intact (PDR 10); all anodes located. Inside the pintle were heavy coverage of acorn barnacles as well as filamentous algae.

Description	Level of fouling (LOF)	PDR Rating	Comments
Top of Rudder	Rank 3 - 6-15% of visible surfaces - Considerable macrofouling on surfaces.	PDR: 20 AF paint missing from edges, corners, seams, welds, rivet or bolt heads to expose anti corrosion (AC) paint. (undercoat).	No additional comments
Pintle Bearing - Port	Rank 3 - 6-15% of visible surfaces - Considerable macrofouling on surfaces.	PDR: 20 AF paint missing from edges, corners, seams, welds, rivet or bolt heads to expose anti corrosion (AC) paint. (undercoat).	No additional comments
Pintle Bearing - Starboard	Rank 3 - 6-15% of visible surfaces - Considerable macrofouling on surfaces.	PDR: 20 AF paint missing from edges, corners, seams, welds, rivet or bolt heads to expose anti corrosion (AC) paint. (undercoat).	No additional comments
Rudder Hinge - Port	Rank 3 - 6-15% of visible surfaces - Considerable macrofouling on surfaces.	PDR: 20 AF paint missing from edges, corners, seams, welds, rivet or bolt heads to expose anti corrosion (AC) paint. (undercoat).	No additional comments
Rudder Hinge - Starboard	Rank 3 - 6-15% of visible surfaces - Considerable macrofouling on surfaces.	PDR: 20 AF paint missing from edges, corners, seams, welds, rivet or bolt heads to expose anti corrosion (AC) paint. (undercoat).	No additional comments
Leading Edge	Rank 3 - 6-15% of visible surfaces - Considerable macrofouling on surfaces.	PDR: 20 AF paint missing from edges, corners, seams, welds, rivet or bolt heads to expose anti corrosion (AC) paint. (undercoat).	No additional comments
Rudder Flat Plate - Port	Rank 3 - 6-15% of visible surfaces - Considerable macrofouling on surfaces.	PDR: 20 AF paint missing from edges, corners, seams, welds, rivet or bolt heads to expose anti corrosion (AC) paint. (undercoat).	No additional comments

Description	Level of fouling (LOF)	PDR Rating	Comments
Rudder Flat Plate - Starboard	Rank 3 - 6-15% of visible surfaces - Considerable macrofouling on surfaces.	PDR: 20 AF paint missing from edges, corners, seams, welds, rivet or bolt heads to expose anti corrosion (AC) paint. (undercoat).	No additional comments
Inspection Port - Port	Rank 3 - 6-15% of visible surfaces - Considerable macrofouling on surfaces.	PDR: 20 AF paint missing from edges, corners, seams, welds, rivet or bolt heads to expose anti corrosion (AC) paint. (undercoat).	No additional comments
Inspection Port - Starboard	Rank 3 - 6-15% of visible surfaces - Considerable macrofouling on surfaces.	PDR: 20 AF paint missing from edges, corners, seams, welds, rivet or bolt heads to expose anti corrosion (AC) paint. (undercoat).	No additional comments
Anodes	Rank 1 - 0% Slime Layer on some or all surfaces. No macrofouling.	PDR: 10 Anti-Foul (AF) Paint intact.	No additional comments

INSPECTOR COMMENTS

The rudder carries the highest fouling level recorded on the vessel. Coating breakdown is limited to edges, seams and points of mechanical movement rather than the main plate, so the antifouling system is still performing across the body — the exposed undercoat is a maintenance flag for the next docking, not an active corrosion concern. Biosecurity risk for this niche is moderate given the developed macrofouling; it should be prioritised for cleaning. No invasive marine species identified.









5.2 PROPELLER

5.2.1 OBSERVATIONS AND FINDINGS

Propeller inspected in approximately 3 m visibility. Boss clean with no fouling (LoF Rank 0). Blades 1 to 5 each showed scattered macrofouling in small patches (LoF Rank 2, 1–5% coverage). No antifouling coating present on the propeller.

Description	Level of fouling (LOF)	PDR Rating	Comments
Boss	Rank 0 - 0% No slime layer. No macrofouling. Only clean surfaces.	No PDR rating allocated	No additional comments
Blade 1	Rank 2 - 1-5% of visible surfaces - Macrofouling present in small patches or a few isolated individuals or small colonies.	No PDR rating allocated	No additional comments
Blade 2	Rank 2 - 1-5% of visible surfaces - Macrofouling present in small patches or a few isolated individuals or small colonies.	No PDR rating allocated	No additional comments
Blade 3	Rank 2 - 1-5% of visible surfaces - Macrofouling present in small patches or a few isolated individuals or small colonies.	No PDR rating allocated	No additional comments
Blade 4	Rank 2 - 1-5% of visible surfaces - Macrofouling present in small patches or a few isolated individuals or small colonies.	No PDR rating allocated	No additional comments
Blade 5	Rank 2 - 1-5% of visible surfaces - Macrofouling present in small patches or a few isolated individuals or small colonies.	No PDR rating allocated	No additional comments
Rope Guard	Rank 4 - 16-40% of visible surfaces - Extensive macrofouling present but more than half of surfaces without biofouling.	PDR: 70 AF/AC paint removed from edges, corners, seams, welds, rivet or bolt heads to expose steel substrate with corrosion present.	No additional comments

INSPECTOR COMMENTS

Light, even fouling across all five blades is consistent with a bare bronze propeller and the expected reaccumulation pattern. The burden is to propulsive efficiency rather than coating integrity; a polish during the next clean will restore performance. Biosecurity risk low. No invasive marine species identified.



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2020-05-25 01:40:07 PROBE DEPTH:2.1M ALT:5.5M TEMP:28C HOG:204 H0E -



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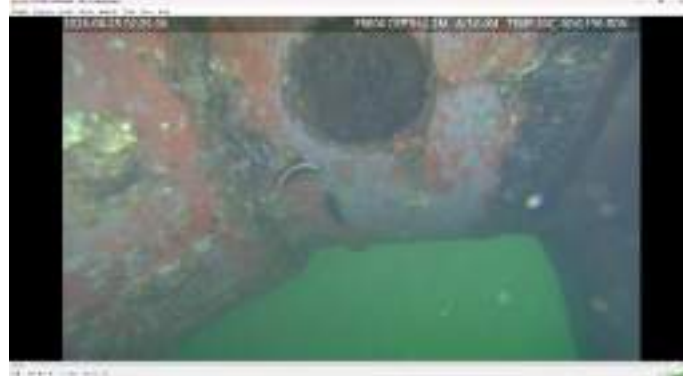


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2020-05-25 01:40:38 PROBE DEPTH:1.1M ALT:5.5M TEMP:28C HOG:174 H0E -







5.3 SEA CHEST 1

5.3.1 OBSERVATIONS AND FINDINGS

External face inspected with light scattered macrofouling (LoF Rank 2, 1–5% coverage) and minor antifouling loss with visible brush swirl marks in the outermost layer (PDR 30). Internal sea chest not visible; internal grating reported as fouled. Internal surfaces recorded clean (Rank 0, PDR 10) on the basis of no access — not directly assessed.

Description	Level of fouling (LOF)	PDR Rating	Comments
External	Rank 2 - 1-5% of visible surfaces - Macrofouling present in small patches or a few isolated individuals or small colonies.	PDR: 30 AF paint missing from slightly curved or flat areas to expose underlying AC paint; An AF paint with visible brush swirl marks within the outermost layer – not extending into underlying layers of paint.	No additional comments
Internal	Rank 0 - 0% No slime layer. No macrofouling. Only clean surfaces.	PDR: 10 Anti-Foul (AF) Paint intact.	internal sea chest not visible

INSPECTOR COMMENTS

External condition is sound with only incipient fouling and cosmetic coating wear. The significant item is the internal space, which could not be inspected and the grating of which was reported fouled — this is the one outstanding biosecurity uncertainty for the vessel and should be confirmed before clearance. No invasive marine species identified on the accessible surfaces.



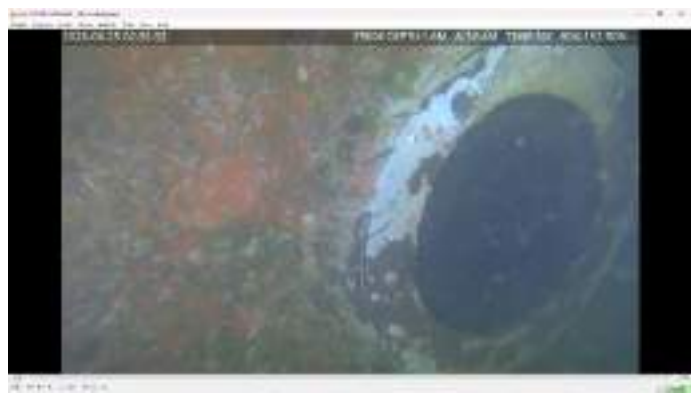
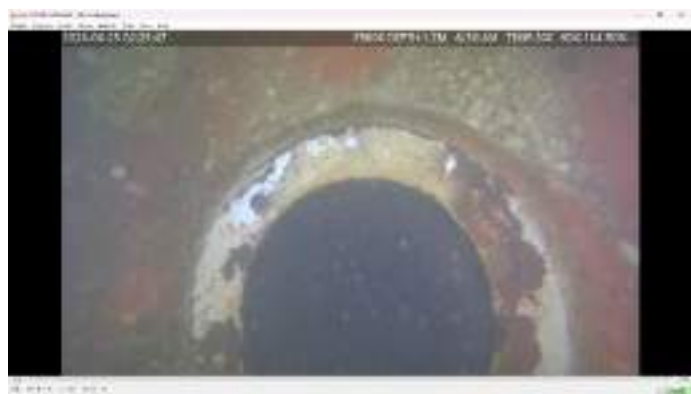


5.4 TRANSDUCERS

5.4.1 OBSERVATIONS AND FINDINGS

clear and cleaned recently

Description	Level of fouling (LOF)	PDR Rating	Comments
No description available	Rank 1 - 0% Slime Layer on some or all surfaces. No macrofouling.	PDR: 40 AF paint missing from intact blisters to expose AC paint or an AF coating with visible brush swirl marks exposing the next underlying layer of AF or AC paint.	No additional comments



6. RECOMMENDATIONS

1. Proceed with targeted in-water cleaning of the rudder and propeller to remove the recorded macrofouling and restore propulsive efficiency.
2. Inspect the internal sea chest spaces and gratings by diver or improved-access ROV pass to confirm fouling extent and IMS status before final clearance.
3. Schedule rudder antifouling repair at the next dry dock opportunity at the edges, seams, fixings and movement points where undercoat is exposed (PDR 20).
4. Establish a 6-month inspection interval to monitor fouling reaccumulation on the running gear.

Please be advised that CCTV imagery and additional still photographs may be provided upon client request.

Thank you for the opportunity to provide our services.

If we can be of any future assistance, please do not hesitate to contact us at any time.

Supervisor Declaration

I, Mat Harvey, declare my inspection and findings outlined above to be true and correct to the best of my knowledge on the date of inspection.



Signed:

Date: Wednesday 24 June 2026

END OF REPORT

Document Status		
Revision:	Document Owner: Adam Falconer-West	Document Number:
Action	Name and Position	Date
Prepared By		Wednesday 24 June 2026
Reviewed By	Mat	-
Approved By		

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